

Software Bug Report

Bug 1: Missing Include Statement

Bug Documentation

Description: The code attempts to use ifstream for file operations but does not include the necessary header.

Steps to Reproduce:

1. Compile the code as provided
2. The compiler will issue an error about undefined reference to ifstream

Expected Behavior: The code should compile without errors.

Actual Behavior: Compilation fails with an error about ifstream being undefined.

Bug Resolution

Root Cause: Code is missing #include <fstream> directive which is required for file I/O operations.

Fix Summary: Added the missing include directive at the top of the file:

cpp

```
#include <fstream> //A Added missing include for ifstream
```

Bug 2: Email Object Initialization Syntax

Bug Documentation

Description: The initialization of the currentEmail object in the EmailManager class uses outdated syntax that might cause issues with some compilers.

Steps to Reproduce:

1. Compile the code with a compiler that enforces stricter C++ standards
2. The compiler might raise warnings or errors about the initialization

Expected Behavior: The Email object should be initialized using modern C++ uniform initialization syntax.

Actual Behavior: The code uses function-style initialization which can be problematic in some contexts and is less consistent with modern C++ practices.

Bug Resolution

Root Cause: The initialization syntax Email("", "", "01-01-2000") is less preferred in modern C++ compared to uniform initialization.

Fix Summary: Changed the initialization syntax to use uniform initialization:

cpp

```
//D Email currentEmail = Email("", "", "01-01-2000");
```

```
Email currentEmail{"", "", "01-01-2000"}; //A Fixed initialization syntax
```

Summary of Changes

- Added 1 missing include directive for `<fstream>`
- Updated 1 object initialization to use modern C++ syntax
- No other functional changes were required as the core logic appeared to be working correctly